

Figure6A

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Intro

Taxonomic diversity loss simulated by increasing water temperature scenarios and comparing semesters to reproduce the results of Figure 6A from the original publication. Seasonal dynamics of the coastal microbiome and its association with environmental factors.

1. Set the environment

```
library(tidyverse)
library(vegan)
source("../scripts/resources/increasing_temp_simulation.R")
```

2. Load data

asvs_workable.tsv contains the rarefied ASV abundance profiles, with samples as rows and ASVs as columns.

samo_metadata_workable.tsv contains the metadata for the samples. date2season2community.tsv is a table mapping the date, season, and community columns.

```
ABUND <- read_tsv("../data/asvs_workable.tsv.gz", show_col_types = FALSE)
METADATA <- read_tsv("../data/samo_metadata_workable.tsv", show_col_types = FALSE)
DATE2SEASON2COMMUNITY <- read_tsv("../data/date2season2community.tsv", show_col_types = FALSE)
```

3. Convert ABUND to long format and map metadata

```
ABUND_long <- ABUND %>%
  as.data.frame() %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(~Date, names_to = "asv_id", values_to = "abund") %>%
  left_join(x = .,
    y = METADATA %>% dplyr::select(Date, sample_name, `Temperature_C`),
    by = c("Date")) %>%
  left_join(x = .,
    y = DATE2SEASON2COMMUNITY ,
    by = "Date") %>%
  filter(abund > 0) # it is critical to remove zeros, other wise asv ranges in S1 and S2 a
```

4. Compute temperature range for each semester

```
semesters2range <- ABUND_long %>%
  group_by(Community) %>%
  summarise(min_temp = min(Temperature_C),
```

```

max_temp = max(Temperature_C))

s12range <- semesters2range %>% filter(Community == "S1")
s22range <- semesters2range %>% filter(Community == "S2")

```

5. Compute temperature range for each ASV

```

asvs2range <- ABUND_long %>%
  group_by(asv_id) %>%
  summarise(min_temp = min(Temperature_C),
            max_temp = max(Temperature_C)) %>%
  rename(seq_id = asv_id)

```

6. Extract communities associated to each semester

```

ABUND <- ABUND %>%
  column_to_rownames(var = "Date")

s1_i <- DATE2SEASON2COMMUNITY %>%
  filter(Community == "S1") %>%
  pull(Date) %>%
  as.character()

ABUND_s1 <- ABUND[s1_i,]

s2_i <- DATE2SEASON2COMMUNITY %>%
  filter(Community == "S2") %>%
  pull(Date) %>%
  as.character()

ABUND_s2 <- ABUND[s2_i,]

# Sanity check
all(colnames(ABUND_s1) == colnames(ABUND_s2))

## [1] TRUE

```

7. Simulate diversity loss in semester S1

```

temp_increase <- seq(0, 3, 0.2)

s1_temp_sim <- increasing_temp_simulation(abund_wide = ABUND_s1,
                                         abund_wide_ref = ABUND_s2,
                                         seqs2range = asvs2range,
                                         semester2range = s12range,
                                         temp_increase = temp_increase)

```

8. Simulate diversity loss in semester S2

```

s2_temp_sim <- increasing_temp_simulation(abund_wide = ABUND_s2,
                                         abund_wide_ref = ABUND_s1,
                                         seqs2range = asvs2range,

```

```
semester2range = s2range,
temp_increase = temp_increase)
```

9. Extract diversity values from nested lists

```
s1_temp_div_df <- map_dfr(s1_temp_sim, ~ {
  tibble(
    temp = .x$temp,
    diversity = .x$div$diversity,
    rich_obs = .x$div$rich_obs,
    rich_chao1 = .x$div$rich_chao1,
    semester = "S1"
  )
})

s2_temp_div_df <- map_dfr(s2_temp_sim, ~ {
  tibble(
    temp = .x$temp,
    diversity = .x$div$diversity,
    rich_obs = .x$div$rich_obs,
    rich_chao1 = .x$div$rich_chao1,
    semester = "S2"
  )
})

temp_div_long_df <- rbind(s1_temp_div_df, s2_temp_div_df) %>%
  pivot_longer(cols = c("diversity", "rich_obs", "rich_chao1"),
    names_to = "metric", values_to = "value") %>%
  group_by(temp, semester, metric) %>%
  summarize(mean = mean(value),
    sd = sd(value))
```

`summarise()` has grouped output by 'temp', 'semester'. You can override using the `.groups` argument

10. Identify significant differences in diversity loss

```
temp_div_long_df2 <- rbind(s1_temp_div_df, s2_temp_div_df) %>%
  pivot_longer(cols = c("diversity", "rich_obs", "rich_chao1"),
    names_to = "metric", values_to = "value")

format_pvalues <- function(anova_welch) {
  p_value <- anova_welch$p.value

  if (is.na(anova_welch$p.value) == T) {
    return(NA)
  } else {
    if (p_value < 0.001) {
      p_value <- "<0.001"
    } else {
      if (p_value < 0.01) {

```

```

    p_value <- "<0.01"
  } else {
    if (p_value < 0.05) {
      p_value <- "<0.05"
    } else {
      p_value <- ">0.05"
    }
  }
}
}
}
return(p_value)
}

for (i in temp_increase[-1]) {

  temp_div_df_i <- temp_div_long_df2 %>%
    filter(temp == i)

  # diversity
  welch_anova_i_div <- temp_div_df_i %>%
    filter(metric == "diversity") %>%
    oneway.test(value ~ semester,
                 data = .,
                 var.equal = FALSE)

  rows_i <- temp_div_long_df$temp == i & temp_div_long_df$metric == "diversity"
  temp_div_long_df[rows_i, "p_value"] <- welch_anova_i_div$p.value
  temp_div_long_df[rows_i, "p_value_formatted"] <- format_pvalues(welch_anova_i_div)

  # richness observed
  welch_anova_i_rich_obs <- temp_div_df_i %>%
    filter(metric == "rich_obs") %>%
    oneway.test(value ~ semester,
                 data = .,
                 var.equal = FALSE)

  rows_i <- temp_div_long_df$temp == i & temp_div_long_df$metric == "rich_obs"
  temp_div_long_df[rows_i, "p_value"] <- welch_anova_i_rich_obs$p.value
  temp_div_long_df[rows_i, "p_value_formatted"] <- format_pvalues(welch_anova_i_rich_obs)

  # richness Chao
  welch_anova_i_rich_chao1 <- temp_div_df_i %>%
    filter(metric == "rich_chao1") %>%
    oneway.test(value ~ semester,
                 data = .,
                 var.equal = FALSE)

  rows_i <- temp_div_long_df$temp == i & temp_div_long_df$metric == "rich_chao1"
  temp_div_long_df[rows_i, "p_value"] <- welch_anova_i_rich_chao1$p.value
  temp_div_long_df[rows_i, "p_value_formatted"] <- format_pvalues(welch_anova_i_rich_chao1)
}

```

```
}
```

11. Plot diversity loss

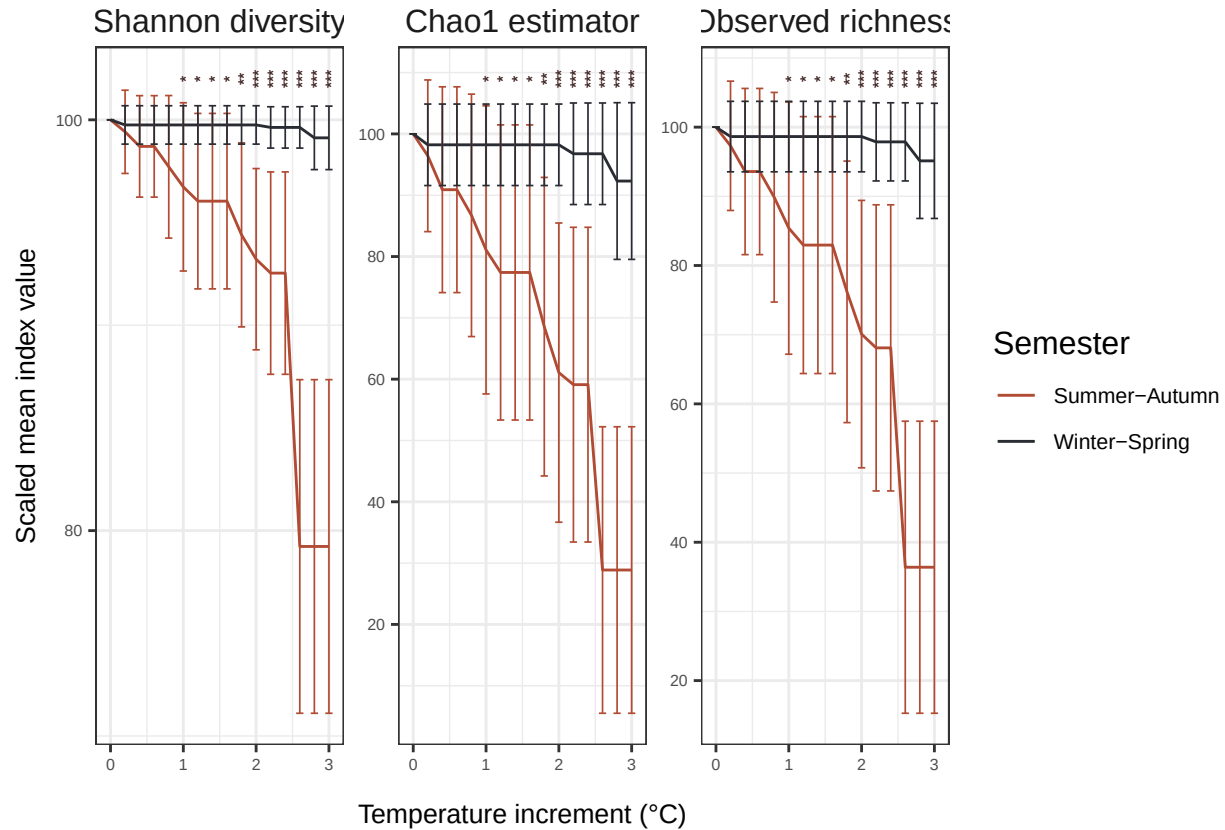
```
text_size <- 8

plot_labels <- c("diversity" = "Shannon diversity",
                "rich_obs" = "Observed richness",
                "rich_chao1" = "Chao1 estimator")

temp_div_long_df <- temp_div_long_df %>%
  mutate(y_max = if_else(metric == "diversity", 102, NA),
         y_max = if_else(metric == "rich_obs", 107, y_max),
         y_max = if_else(metric == "rich_chao1", 109, y_max))

div_plot <- ggplot(temp_div_long_df, aes(x = temp, y = mean, color = semester)) +
  geom_line() +
  facet_wrap(~metric, scales = "free_y", labeller = as_labeller(plot_labels)) +
  theme_minimal() +
  xlab("Temperature increment (°C)") +
  ylab("Scaled mean index value") +
  scale_y_continuous(breaks = seq(0, 100, 20)) +
  scale_color_manual(values = c("#b14b34", "#292c33"),
                    labels = c("S1" = "Summer-Autumn", "S2" = "Winter-Spring"),
                    name = "Semester") +
  geom_errorbar(aes(ymin = mean - sd, ymax = mean + sd), linewidth = 0.3, width = 0.1) +
  theme_bw() +
  theme(
    axis.text.x = element_text(size = text_size-2),
    axis.text.y = element_text(size = text_size-2),
    axis.title.x = element_text(size = text_size +2, margin = unit(c(4,0,0,0), "mm")),
    axis.title.y = element_text(size = text_size +2, margin = unit(c(0,2,0,0), "mm")),
    strip.text = element_text(size = text_size+4),
    strip.background = element_blank(),
    # increase legend text
    legend.text = element_text(size = text_size),
    legend.title = element_text(size = text_size+4)
  ) +
  geom_text(data = temp_div_long_df %>% filter(p_value_formatted == "<0.001"),
            aes(x = temp, y = y_max, label = "***", vjust = 0.7),
            size = 2, angle = 90, show.legend = FALSE) +
  geom_text(data = temp_div_long_df %>% filter(p_value_formatted == "<0.01"),
            aes(x = temp, y = y_max, label = "**", vjust = 0.7),
            size = 2, angle = 90, show.legend = FALSE) +
  geom_text(data = temp_div_long_df %>% filter(p_value_formatted == "<0.05"),
            aes(x = temp, y = y_max, label = "*", vjust = 0.7),
            size = 2, angle = 90, show.legend = FALSE)

div_plot
```



12. Print session info

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 20.04.6 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.9.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.9.0
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C               LC_TIME=en_US.UTF-8       LC_COLLATE=en_US.UTF-8
## [6] LC_MESSAGES=en_US.UTF-8   LC_PAPER=en_US.UTF-8       LC_NAME=C                  LC_ADDRESS=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] fANCOVA_0.6-1      maps_3.4.2          corrgram_1.14       ggpubr_0.6.0        ggcorrplot_0.1.4.1
## [8] permute_0.9-7      lubridate_1.9.3     forcats_1.0.0       stringr_1.5.1       dplyr_1.1.4
```

```
## [15] tidyr_1.3.1      tibble_3.2.1      ggplot2_3.5.1      tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.5      xfun_0.48          rstatix_0.7.2      tzdb_0.4.0         vctrs_0.6.5        tools_
## [9] fansi_1.0.6       highr_0.11         cluster_2.1.6      pkgconfig_2.0.3    Matrix_1.7-0       lifecycle_
## [17] tinytex_0.53      munsell_0.5.1      carData_3.0-5      htmltools_0.5.8.1  yaml_2.3.10        Formula_
## [25] car_3.1-3         MASS_7.3-61        abind_1.4-8        nlme_3.1-166       tidyselect_1.2.1   digest_
## [33] splines_4.4.2     cowplot_1.1.3      fastmap_1.2.0      colorspace_2.1-1   cli_3.6.3          magrittr_
## [41] withr_3.0.1       scales_1.3.0       backports_1.5.0    bit64_4.5.2        timechange_0.3.0   rmarkdown_
## [49] hms_1.1.3         evaluate_1.0.1     knitr_1.48         mgcv_1.9-1         rlang_1.1.4        glue_
## [57] R6_2.5.1
```